

CT-QMC with Complex Hybridization Function

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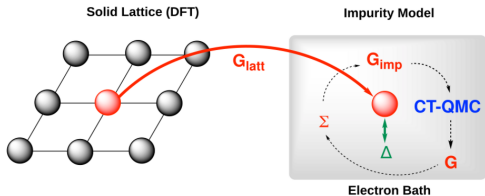
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Dynamical Mean-Field Theory (DFT+DMFT)

→ Mapping the periodic many-body problem into a many-body impurity model d'Anderson



Kotliar *et al. Rev. Mod. Phys.* **2006**, 78,865

$$\begin{aligned}
 H &= H^{\text{Bath}} + H^{\text{Impurity}} + H^{\text{Hybridization}} \\
 &= \sum_{l,\sigma} \epsilon_l c_{l,\sigma}^\dagger c_{l,\sigma} + \sum_{i,\sigma} \epsilon_i d_{i,\sigma}^\dagger d_{i,\sigma} + H^{\text{Int}} \\
 &\quad + \sum_{p,i,\sigma} V_p (c_{p,\sigma}^\dagger d_{i,\sigma} + d_{i,\sigma}^\dagger c_{p,\sigma})
 \end{aligned}$$

■ Interacting Hamiltonian

■ Density-Density

$$\hat{H}^{\text{Int}} = \frac{1}{2} \sum_{i,j,\sigma} [U \hat{n}_{i\sigma} \hat{n}_{j\bar{\sigma}} + (U - J) \hat{n}_{i\sigma} \hat{n}_{j\sigma}]$$



■ Rotationnally Invariant

$$\hat{H}^{\text{Int}} = \frac{1}{2} \sum_{i,j,k,l,\sigma,\bar{\sigma}} U_{ijkl} c_{i\sigma}^\dagger c_{j\bar{\sigma}}^\dagger c_{k\bar{\sigma}} c_{l\sigma}$$



■ Impurity model solved by Monte-Carlo

■ CT-QMC (Hybridization expansion)

Continuous-time QMC

- Split the single impurity Hamiltonian

$$H^{\text{SIAM}} = \underbrace{\sum_{p,\sigma} \epsilon_p c_{p,\sigma}^\dagger c_{p,\sigma} + \sum_{i,\sigma} \epsilon_i d_{i,\sigma}^\dagger d_{i,\sigma} + H^{\text{Int}}}_{H^{\text{local}}} + \underbrace{\sum_{p,i,\sigma} V_p (c_{p,\sigma}^\dagger d_{i,\sigma} + d_{i,\sigma}^\dagger c_{p,\sigma})}_{H^{\text{Hyb}}}$$

- Rewrite the partition function

$$Z = \text{Tr} \left[e^{-\beta H^{\text{SIAM}}} \right] = \text{Tr} \left[e^{-\beta H^{\text{loc}}} \mathcal{T} e^{-\int_0^\beta d\tau H^{\text{Hyb}}(\tau)} \right]$$

- Hybridization expansion in time ordered exponentials

$$Z = \sum_{n=0}^{\infty} \int_0^\beta d\tau_1 \dots \int_{\tau_{n-1}}^\beta d\tau_n \int_0^\beta d\tau'_1 \dots \int_{\tau'_{n-1}}^\beta d\tau'_n \\ \times \text{Tr} \left[e^{-\beta H^{\text{loc}}} \mathcal{T} H_1^{\text{Hyb}}(\tau_n) H_2^{\text{Hyb}}(\tau'_n) \dots H_1^{\text{Hyb}}(\tau_1) H_2^{\text{Hyb}}(\tau'_1) \right]$$

Continuous-time QMC

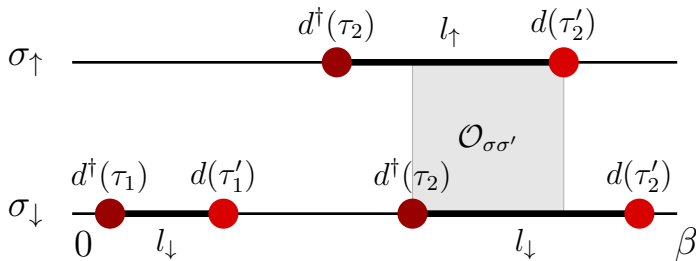
Hybridization expansion

$$\begin{aligned}
 Z &= Z_{\text{bath}} \sum_{n_\sigma} \prod_{\sigma} \int_0^\beta d\tau_1^\sigma \dots \int_{\tau_{n_\sigma}^\sigma - 1}^\beta d\tau_{n_\sigma}^\sigma \int_0^\beta d\tau_1'^\sigma \dots \int_{\tau_{n_\sigma}^\sigma - 1}^\beta d\tau_{n_\sigma}^{\prime\sigma} \\
 &\quad \times \text{Tr}_d \left(e^{-\beta H_{\text{loc}}} \mathcal{T} \prod_{\sigma} d_{\sigma}(\tau_{n_\sigma}^\sigma) d_{\sigma}^{\dagger}(\tau_{n_\sigma}^{\prime\sigma}) \dots d_{\sigma}(\tau_1^\sigma) d_{\sigma}^{\dagger}(\tau_1^{\prime\sigma}) \right) \\
 &\quad \times \frac{1}{Z_{\text{bath}}} \text{Tr}_c \left(e^{-\beta H_{\text{bath}}} \mathcal{T} \prod_{\sigma} \sum_{p_1 \dots p_{n_\sigma}} \sum_{p'_1 \dots p'_{n_\sigma}} V_{p_1 \sigma}^* V_{p'_1 \sigma} \dots V_{p_{n_\sigma} \sigma}^* V_{p'_{n_\sigma} \sigma} \right. \\
 &\quad \left. c_{p_{n_\sigma}}^{\dagger}(\tau_{n_\sigma}^\sigma) c_{p'_{n_\sigma}}(\tau_{n_\sigma}^{\prime\sigma}) \dots c_{p_1 \sigma}^{\dagger}(\tau_1^\sigma) c_{p'_1 \sigma}(\tau_1^{\prime\sigma}) \right) \\
 &= Z_{\text{bath}} \times \underbrace{\sum_c \omega_{\text{loc}}^c}_{\text{Impurity}} \times \underbrace{\sum_c \omega_{\text{bath}}^c}_{\text{Bath}}
 \end{aligned}$$

CT-QMC Hybridization Expansion

- Trace on the impurity (Density-Density interaction with segment solver)

$$\begin{aligned}\omega_{\text{loc}} &= \text{Tr}_{\text{d}} \left(e^{-\beta H_{\text{loc}}} \mathcal{T} \prod_{\sigma} d_{\sigma}(\tau_{n_{\sigma}}^{\sigma}) d_{\sigma}^{\dagger}(\tau_{n_{\sigma}}^{\prime\sigma}) \dots d_{\sigma}(\tau_1^{\sigma}) d_{\sigma}^{\dagger}(\tau_1^{\prime\sigma}) \right) \\ &= \exp \left(\mu \sum_{\sigma} (l_{\sigma\uparrow} + l_{\sigma\downarrow}) - \sum_{\sigma \neq \sigma'} U \times \mathcal{O}_{\sigma\sigma'} \right)\end{aligned}$$



CT-QMC Hybridization Expansion

- Trace on the bath (Density-Density interaction with segment solver)

$$\begin{aligned}\omega_{\text{bath}} &= \frac{1}{Z_{\text{bath}}} \text{Tr}_c \left(e^{-\beta H_{\text{bath}}} \mathcal{T} \prod_{\sigma} \sum_{p_1 \dots p_{n\sigma}} \sum_{p'_1 \dots p'_{n\sigma}} V_{p_1\sigma}^* V_{p'_1\sigma} \dots V_{p_{n\sigma}}^* V_{p'_{n\sigma}} \right. \\ &\quad \left. c_{p_{n\sigma}}^{\dagger}(\tau_{n\sigma}^{\sigma}) c_{p'_{n\sigma}}(\tau'_{n\sigma}) \dots c_{p_1\sigma}^{\dagger}(\tau_1^{\sigma}) c_{p'_1\sigma}(\tau'_1) \right) \\ &= \prod_{\sigma} \det \Delta^{\sigma}(\tau' - \tau)\end{aligned}$$

with $\Delta^{\sigma}(\tau' - \tau)$ the hybridization function defined as

$$\Delta^{\sigma}(\tau' - \tau) = \sum_p \frac{V_{p\sigma} V_{p\sigma}^*}{1 + e^{-\beta \epsilon_p}} \begin{cases} -e^{-\epsilon_p(\tau' - \tau)} & \tau' - \tau > 0 \\ e^{-\epsilon_p(\beta + (\tau' - \tau))} & \tau' - \tau < 0 \end{cases}$$

CT-QMC Hybridization Expansion

- Density-Density interaction segment solver in Abinit : a 3rd possibility

$$Z = Z_{\text{bath}} \times \sum_c \omega_{\text{loc}}^c \times \sum_c \omega_{\text{bath}}^c$$

- Real and diagonal hybridization function
→ *dmft_solv* = 5 ("Diag") (no SOC / *d*-elements)
- Real and offdiagonal hybridization function
→ *dmft_solv* = 8 ("OffDiag") (SOC / cubic *f*-elements)
- Complex and offdiagonal hybridization function
→ *dmft_solv* = 10 ("ImOffDiag") (SOC / low-symmetry *f*-elements)

Practical Aspects

- Inverse Fourier transform to get $\Delta(\tau)$

$$\begin{aligned}\Delta_{ab}(\tau) &= \frac{1}{\beta} \sum_{n=-\infty}^{\infty} \Delta_{ab}(i\omega_n) e^{-i\omega_n \tau} \\ &= \frac{1}{\beta} \left(\sum_{n=0}^{\infty} \left[\Delta_{ab}(i\omega_n) - \frac{A}{i\omega_n} \right] e^{-i\omega_n \tau} + \sum_{n=0}^{\infty} \left[\Delta_{ab}^*(i\omega_n) + \frac{A}{i\omega_n} \right] e^{i\omega_n \tau} \right) - \frac{A}{2}\end{aligned}$$

with A the asymptotic limit to ensure $\lim_{n \rightarrow \infty} \Delta(i\omega_n) = \frac{-A}{i\omega_n}$ (DMFT_HYBRI_LIMIT)

Shinaoka et al. Comput. Phys. Commun. 2017, 215, 128

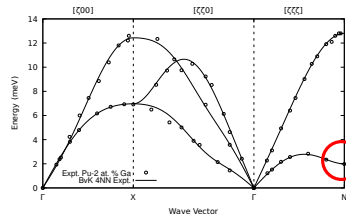
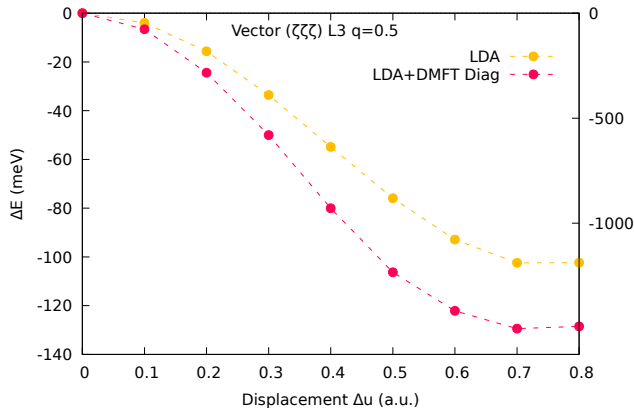
- Fourier transform to get $G(\tau)$

$$\begin{aligned}G(i\omega_n) &= \int_0^{\beta} G(\tau) e^{i\omega_n \tau} d\tau \\ &= \frac{C1}{i\omega_n} + \frac{C2}{(i\omega_n)^2} + \frac{C3}{(i\omega_n)^3} + \frac{C4}{(i\omega_n)^4} + \int_0^{\beta} \frac{G^{(4)}(\tau) e^{i\omega \tau}}{(i\omega_n)^4}\end{aligned}$$

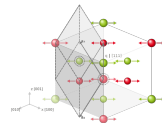
Oudovenko et al. Phys. Rev. B. 2006, 73, 035120

Application : Frozen Phonons of δ -Pu

- Comparison to DFT with $U = 0$

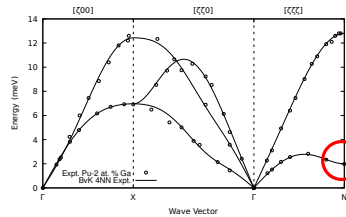
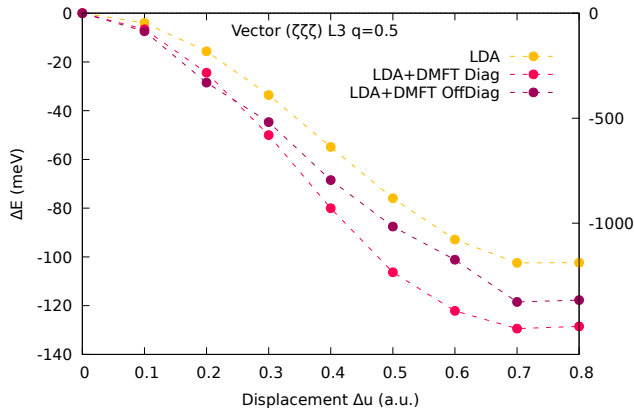


Transverse mode $q=[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$



Application : Frozen Phonons of δ -Pu

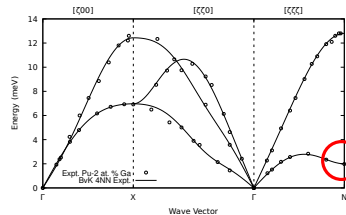
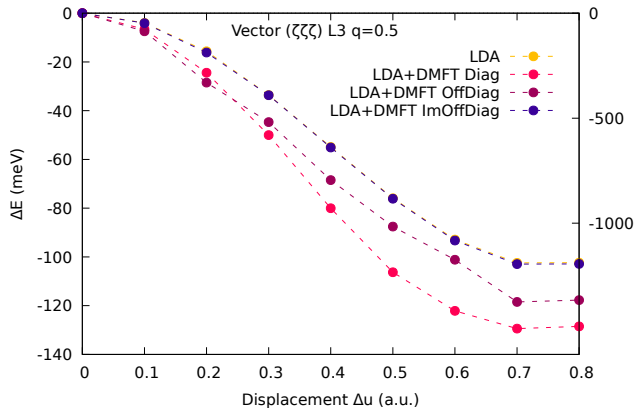
- Comparison to DFT with $U = 0$



Transverse mode $q=[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$

Application : Frozen Phonons of δ -Pu

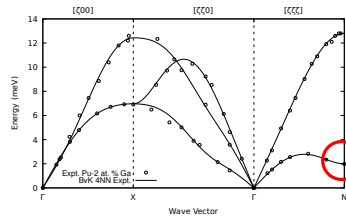
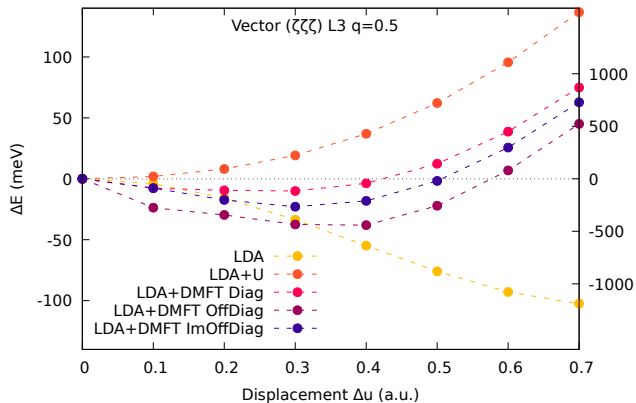
- Comparison to DFT with $U = 0$



Transverse mode $q=[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$

Application : Frozen Phonons of δ -Pu

- Comparison to DFT with $U = 4$ eV



Transverse mode $q=[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}]$

Application : Frozen Phonons of δ -Pu

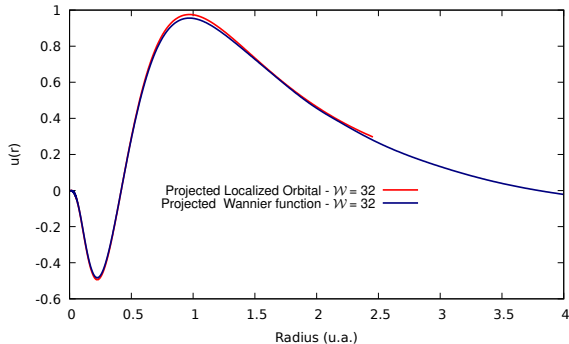
- Influence of localized orbitals
 - Projected localized orbitals

$$|\tilde{\chi}_{\mathbf{k}m}^{\mathbf{R}}\rangle \equiv \sum_{\nu \in \mathcal{W}} \langle \Psi_{\mathbf{k}\nu} | \chi_{\mathbf{k}m}^{\mathbf{R}} \rangle | \Psi_{\mathbf{k}\nu} \rangle$$

followed by orthonormalization

$$|\tilde{w}_{\mathbf{k}m}^{\mathbf{R}}\rangle = \sum_{\mathbf{R}'m'} \left\{ [O(\mathbf{k})]^{-1/2} \right\}_{mm'}^{\mathbf{R}\mathbf{R}'} |\tilde{\chi}_{\mathbf{k}m'}^{\mathbf{R}'}\rangle$$

- Wannier functions
Same as before but we reuse an averaged $|\tilde{w}_{\mathbf{k}m}^{\mathbf{R}}\rangle$ as $\chi_{\mathbf{k}m}^{\mathbf{R}}$ and keep it fix



Application : Frozen Phonons of δ -Pu

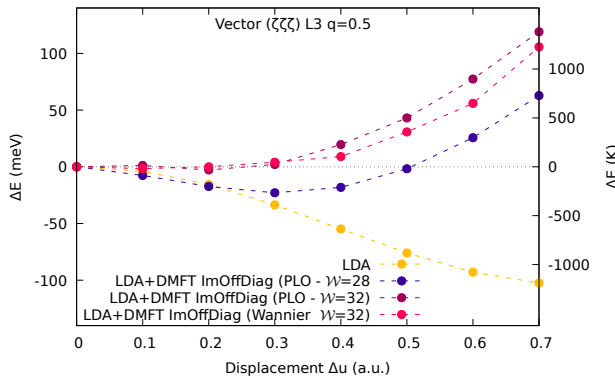
- Influence of localized orbitals
 - Projected localized orbitals

$$|\tilde{\chi}_{\mathbf{k}m}^{\mathbf{R}}\rangle \equiv \sum_{\nu \in \mathcal{W}} \langle \Psi_{\mathbf{k}\nu} | \chi_{\mathbf{k}m}^{\mathbf{R}} \rangle | \Psi_{\mathbf{k}\nu} \rangle$$

followed by orthonormalization

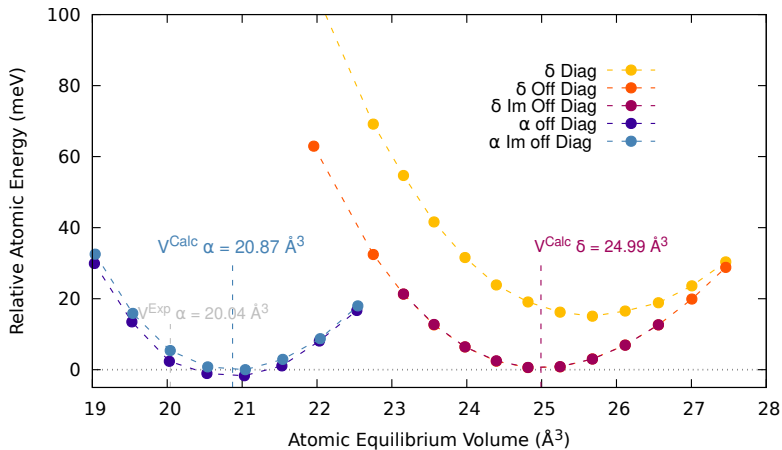
$$|\tilde{w}_{\mathbf{k}m}^{\mathbf{R}}\rangle = \sum_{\mathbf{R}'m'} \left\{ [O(\mathbf{k})]^{-1/2} \right\}_{mm'}^{\mathbf{R}\mathbf{R}'} |\tilde{\chi}_{\mathbf{k}m'}^{\mathbf{R}'}\rangle$$

- Wannier functions
Same as before but we reuse an averaged $|\tilde{w}_{\mathbf{k}m}^{\mathbf{R}}\rangle$ as $\chi_{\mathbf{k}m}^{\mathbf{R}}$ and keep it fix



Application : Atomic Volumes

- Comparison to DFT with $U = 4$ eV



Conclusions

- DFT+DMFT in Abinit
 - Complex CT-QMC segment solver
(DMFT_SOLV = 10,DMFT_HYBRI_LIMIT)
 - Wannier functions as correlated orbitals
(DMFT_PRTWAN,DMFT_ORBITAL,DMFT_RAD)
 - New tutorial on DFT+DMFT with TRIQS/CT-Hyb
 - Local magnetic susceptibility
(DMFT_CTQMC_LOCALPROP)
- Next steps
 - Moments of Green's function
 - Symmetries for supercell